Analytical Cume

February 7, 1998

Identify a *peer-reviewed* journal article on an *analytical chemistry topic* that was published in the last year (Feb. 1997-Feb. 1998). Do *NOT* choose a review article.

Examples of journals containing articles in the area of *analytical chemistry* are:

Analytica Chimica Acta
Analytical Chemistry
Applied Spectroscopy
Chemical Physics Letters
Journal of Chemical Physics
Journal of Chromatography
Journal of Electroanalytical Chemistry
Journal of Physical Chemistry
Journal of the American Chemical Society
Journal of the Electrochemical Society
Langmuir
Sensors and Actuators
Surface Science
Talanta

In reference to your chosen article, answer the following questions with as much detail as possible.

(10 pts) 1. What is the title of the article?
Name the authors (at least identify the principle investigator)?
What is their location(s) (company or university, etc.)?
What is the journal name and year? (Include volume and pages if you remember them.)

(10 pts) 2. What is the take-home message of the article (i.e. key conclusion(s))?

(20 pts) 3. What analytical techniques were used and how do they work? Name instrumental components and physical chemical phenomena that allow the techniques to analyze samples. (How does the technique detect the analyte or sample?)

(15 pts) 4. Describe the main experiment(s) and results.

(15 pts) 5. What is the impact of the article on the scientific community, if any? Why?

6. Imagine that the original manuscript was just sent to you for review, to determine whether it should be published or not.

(a) (5 pts) Choose one of the following regarding the manuscript.
____represents a major development and should be given priority treatment
____can be published without change
____should be published with minor change
____should be published with major change
____should not be published in this journal

(b) (5 pts) Place an "X" in one column for each category. (Reproduce the table in your blue book.)

	no change	**minor change**	**major change**
scientific content			
grammar			
title			
introduction			
summary			
references			
tables			
figures			

(c) (20 pts) Comment on the scientific quality of the article. Do the data support the conclusions?